Rui Chen

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Availability: from mid-2027 | Funding: university scholarship or CSC (both routes open)

RESEARCH INTERESTS

Medicinal chemist combining structure-based design with synthesis and process development, with three years of industrial experience in antiviral drug discovery. That experience built the toolkit modern medicinal chemistry programs require: complex multistep synthesis, conjugate and linker construction, and developability judgment (prodrugs, solid-form control, freedom-to-operate).

EDUCATION

China Pharmaceutical University (CPU) Sep. 2020 – Jun. 2023
M.Sc. in Pharmacy, GPA: 4.00 / 4.00 (88.54/100.00), Top 10%
Shandong University of Traditional Chinese Medicine (SDUTCM) Sep. 2016 – Jun. 2020
B.Sc. in Pharmaceutical Engineering, GPA: 82.48/100, Top 3% (Ranked 7/247)

PUBLICATIONS

6 published papers, including 1 co-first author (Eur. J. Med. Chem.).

Hu, T.[1]; Huang, J.[1]; Chen, R.[1]; Zhang, H.; Liu, M.; Wang, R.; Zhou, W.; Huang, D.; Cao, M.; Li, D.; Li, Z.; Wu, H.; Bian, J. Discovery of CLKs Inhibitors for the Treatment of Non-Small Cell Lung Cancer. Eur. J. Med. Chem. 2024, 280, 116952. doi: 10.1016/j.ejmech.2024.116952 (Joint first author; one of three)

Zhou, L.; Yu, B.; Gao, M.; Chen, R.; Li, Z.; Gu, Y.; Bian, J.; Ma, Y. DNA Framework-Engineered Chimeras Platform Enables Selectively Targeted Protein Degradation. Nat. Commun. 2023, 14 (1), 4510. doi: 10.1038/s41467-023-40244-7

Wu, T.; Yu, B.; Xu, Y.; Du, Z.; Zhang, Z.; Wang, Y.; Chen, H.; Zhang, L.; Chen, R.; Ma, F.; Gong, W.; Yu, S.; Qiu, Z.; Wu, H.; Xu, X.; Wang, J.; Li, Z.; Bian, J. Discovery of Selective and Potent Macrocyclic CDK9 Inhibitors for Osimertinib-Resistant Non-Small-Cell Lung Cancer. J. Med. Chem. 2023, 66 (22), 15340–15361. doi: 10.1021/acs.jmedchem.3c01400

Wu, T.; Wu, X.; Xu, Y.; Chen, R.; Wang, J.; Li, Z.; Bian, J. A Patent Review of Selective CDK9 Inhibitors in Treating Cancer. Expert Opin. Ther. Pat. 2023, 33 (4), 309–322. doi: 10.1080/13543776.2023.2208747

Wang, D.; Li, X.; Gong, G.; Lu, Y.; Guo, Z.; Chen, R.; Huang, H.; Li, Z.; Bian, J. An Updated Patent Review of Glutaminase Inhibitors (2019–2022). Expert Opin. Ther. Pat. 2023, 33 (1), 17–28. doi: 10.1080/13543776.2023.2173573

Xie, Y.; Tian, Y.; Zhang, Y.; Zhang, Z.; Chen, R.; Li, M.; Tang, J.; Bian, J.; Li, Z.; Xu, X. Overview of the Development of Selective Androgen Receptor Modulators (SARMs) as Pharmacological Treatment for Osteoporosis (1998–2021). Eur. J. Med. Chem. 2022, 230, 114119. doi: 10.1016/j.ejmech.2022.114119

PATENTS

Two Chinese invention patent applications on processes for the preparation of an optically active intermediate of a polycyclic carbamoyl pyridone derivative: (I) an asymmetric route; (II) a route avoiding hazardous reagents. Drafted and finalized with IP counsel (2026); filing pending.

RESEARCH

China Pharmaceutical University | Supervisor: Prof. Jinlei Bian

Keywords: PROTAC, molecular glues, linker design, CLK2, CADD, process chemistry

DEGRADER & NEW-MODALITY CHEMISTRY

(1) DNA-linker PROTAC (CDK9 warhead) Apr. 2022 – Jun. 2023

- Responsible for the synthesis, purification and characterization of the ligands the platform is built on (CDK9 and CDK6 warheads, CRBN recruiter).
- Contributed to the study showing that caged DNA fixes ligand spacing and supports efficient protein degradation.
- Co-authored the resulting article, published in Nat. Commun. (2023).

TARGET-BASED DESIGN AND SYNTHESIS

(2) ASCT2 (SLC1A5) inhibitors (Project Leader) Oct. 2021 – Jun. 2023

- Led a virtual screening and activity-assay workflow combining molecular dynamics simulation, docking and pharmacophore screening.
- Analyzed the ligand–target conformational relationship from the crystal structure, then designed and synthesized 31 novel inhibitors.
- Identified a lead with antiproliferative IC_{50} = 0.39 μM in tumor cells, outperforming the reference ASCT2 inhibitor V9302 in the same proliferation assay; the scaffold was newly discovered by virtual screening rather than derived from known chemotypes.
- Completed the M.Sc. thesis on this programme.
- Mentored two undergraduates through the full review-writing process — literature and patent searching, critical reading, and manuscript structure — for a patent review on glutaminase inhibitors (Expert Opin. Ther. Pat. 2023).

(3) CLK2 inhibitors Jun. 2022 – Jun. 2023

- Structure-based optimisation of a CLK2 inhibitor series: the lead reached CLK2 IC50 = 3.9 nM with cellular potency of 0.13 μM in H1299 cells (a 19-fold improvement), and 77 of 83 kinases were inhibited by less than 50% in a panel.

- Performed molecular docking and prepared the docking figures for a co-authored patent review on selective CDK9 inhibitors (Expert Opin. Ther. Pat. 2023).
- Published the CLK2 work in Eur. J. Med. Chem. (2024).

(4) DON ester prodrugs Feb. 2022 – Jun. 2022
- Designed and synthesized a series of DON ester prodrugs.
- Mitigated the intrinsic toxicity of DON and improved its plasma stability and tolerability.

(5) LAT1 inhibitors Aug. 2021 – May. 2022
- Contributed molecular docking and scaffold design to the LAT1 inhibitor program.

PROCESS CHEMISTRY & DEVELOPABILITY

(6) Process chemistry for a novel JAK inhibitor Apr. 2021 – Jan. 2022
- Optimized process and recrystallization parameters, improving the overall yield and enabling rapid synthesis at kilogram scale.

(7) Reference standards for psychotropic drugs Oct. 2020 – Mar. 2021
- Synthesized standardized reference materials for several psychotropic drugs and authored the project closure report (laboratory project for the National Narcotics Control Bureau).

WORK EXPERIENCE

Jiangsu Aidea Pharmaceutical Co., Ltd. | Medicinal Chemist Jul. 2023 – Present

Focus: HIV antivirals and process chemistry — discovery, analogue design and scale-up

(1) Process chemistry for a chiral intermediate
- Developed two alternative synthetic routes to an optically active intermediate of a polycyclic carbamoyl pyridone derivative, targeting lower cost, simpler chiral resolution, and reduced reliance on hazardous reagents.
- Redesigned the chiral resolution step, improving isolation efficiency and eliminating a chromatography-dependent operation not scalable in manufacturing.
- Authored two patent applications on the two routes (drafted and finalized with IP counsel; filing pending).

(2) HIV-1 integrase inhibitor analogue series (ACC017 follow-up)
- Synthesized and characterized an analogue series built on the clinical-stage integrase inhibitor ACC017, targeting improved metabolic stability, oral bioavailability and half-life.
- Achieved a substantial half-life gain in the series; the lead profile fell outside the program's strategic target, so the series was not advanced to in-depth evaluation.
- Contributed to resolving an aqueous-solubility liability and to identifying a developable crystal form for scale-up.

(3) Long-acting injectable prodrug platform
- Designed and synthesized high-molecular-weight prodrug candidates (MW > 1500) to extend the dosing interval and improve bioavailability.
- Evaluated promoiety and linker strategies to balance chemical stability, release kinetics and tolerability.
- Built multistep synthetic routes to complex high-MW conjugates requiring careful control of coupling and purification.

(4) Novel HIV-1 capsid inhibitor – structure-based design
- Designed novel HIV-1 capsid inhibitors using molecular dynamics simulation and docking.
- Synthesized complex polycyclic candidates requiring multistep chiral routes, and evaluated their biological activity.
- Screened intermediate processes, optimized crystallization and reduced impurity formation at scale.

(5) ACC007 – production process improvement
- Redesigned the synthetic route to reduce production cost, and optimized process parameters to cut operational waste at scale.

PROFESSIONAL SKILLS

CADD & AIDD: structure-based design from virtual screening and MD through to synthesis (Schrödinger, MOE, Discovery Studio, AutoDock, GROMACS, PyMOL); completed a full design–synthesis–assay cycle against ASCT2 (31 compounds).

AI & data: apply LLM-based tools to literature mining, patent analysis and AI-assisted design; use AI for SAR reasoning, ADMET triage and docking/MD interpretation.

Experimental: independently design experiments and deliver target compounds, including complex polycyclic and chiral targets; TLC, recrystallization, column chromatography, and structure elucidation by ^{1}H NMR, MS and HPLC.

Software: Office, Zotero, Adobe Illustrator, GraphPad, ChemDraw, MestReNova.

Language: Chinese Mandarin (Level II, Grade A); English: IELTS 6.5 (no band below 6.0).

HONORS & AWARDS

RESEARCH

Bronze Award (Team lead, four-person team), AI for Science Track, 2nd Global Digital-Intelligence Education Innovation Competition (Peking University) Nov. 2025

ACADEMIC

CPU Scholarship, Second Prize (2021 & 2022); CPU First Prize for Freshman Graduate Students (Top 5%) 2020–2022

Shandong Province Outstanding Undergraduate Student Award (Top 1%) Jun. 2020

Provincial Silver Award, "Internet+" Innovation and Entrepreneurship Competition Sep. 2019